

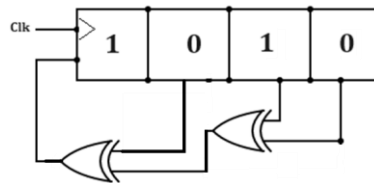
BITS ID: _____ **Name:** _____

Q1 Realize the Boolean expression using the minimum number of Transmission Gates (TGs). [5]

$$F(A, B, C, D) = \sum m(2, 7, 9, 12) + d(5, 6, 8, 10, 13, 14, 15).$$

True and complementary inputs are available for design. **Use C and D as select lines.**

Q2 A 4-bit shift register is shown in the figure below. It is clocked, and with each clock pulse, the pattern gets shifted by a one-bit position to the right. [5]



Assume that the initial value of the shift registers as 1010. Write the content of the shift registers after the following clock pulses:

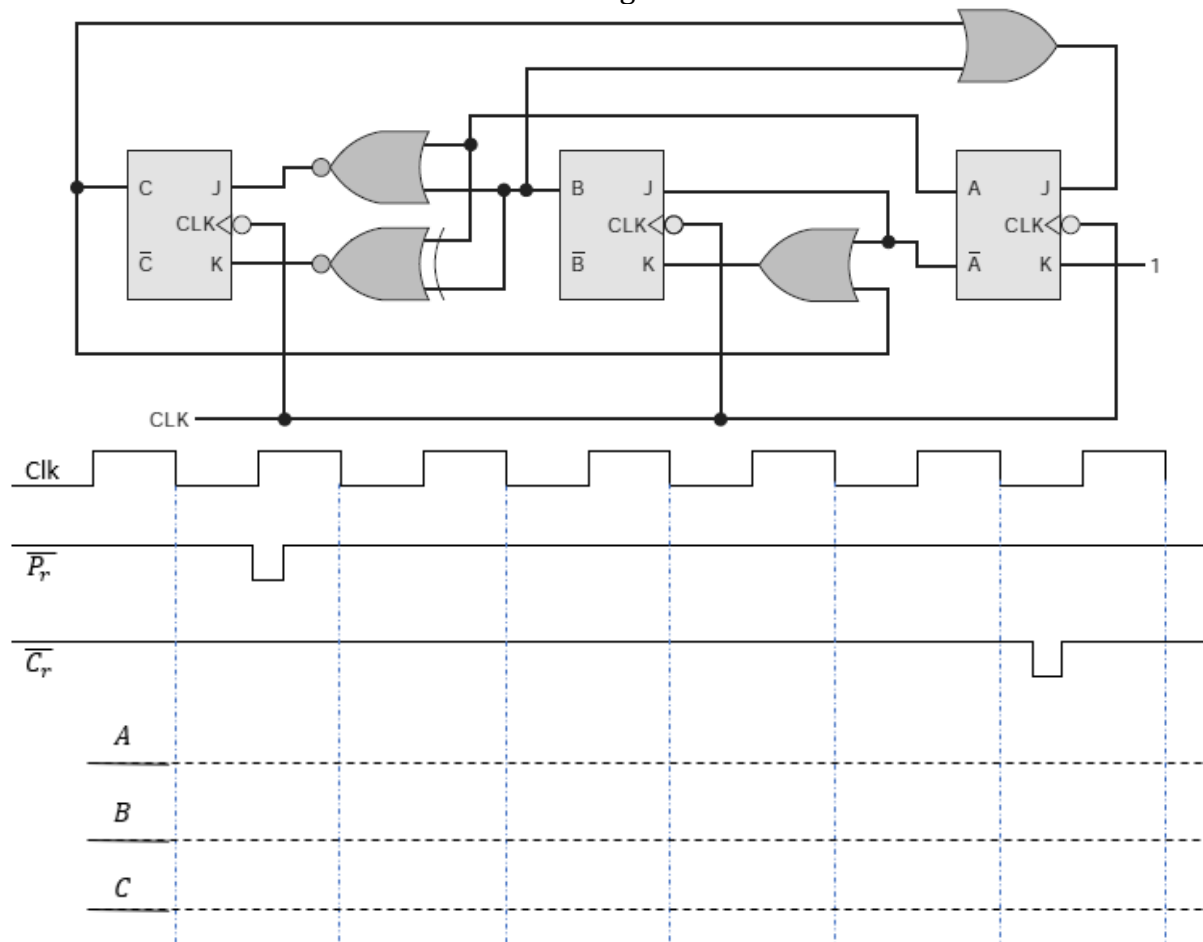
	1	0	1	0
After 1 st clock pulse				
After 5 th clock pulse				
After 9 th clock pulse				
After 11 th clock pulse				

To what values should the shift register be initialized so that the pattern (**1010**) occurs after the first clock pulse:

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Q3	Realize the Boolean function $F(a, b, c) = \sum m(0, 2, 5, 7)$ using • ONE 3-to-8-line decoder with active low output and active low enable • Minimum number of 2-input NAND gates	[5]				
Q4	Realize $F(A, B, C, D) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 10)$ using a minimum number of CMOS transistors. Only true inputs are available. SOP Expression: _____ <table><tr><th>Load Part/Pull Up network</th><th>Driver Part/Pull Down network</th></tr><tr><td></td><td></td></tr></table>	Load Part/Pull Up network	Driver Part/Pull Down network			[5]
Load Part/Pull Up network	Driver Part/Pull Down network					

Q5	Draw the waveform corresponding to the circuit below; assume that preset and clear (both active low) of all the flipflops are commonly connected to " $\overline{P_r}$ " and " $\overline{C_r}$ " respectively. Assume $A = B = C = 0$ before the first active edge.	[5]
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Q6	Realize the following Boolean functions with a minimum size PLA (only true outputs) and complete the PLA programming table.	[5]
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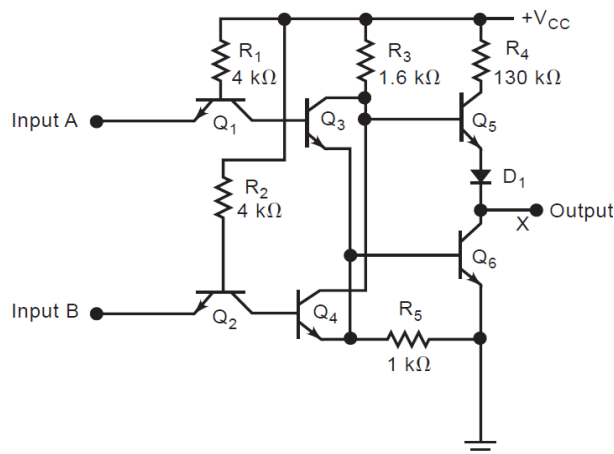
$$F1(A, B, C, D) = \sum m(1, 3, 5, 7, 8, 10, 11) \quad F2(A, B, C, D) = \sum m(6, 7, 8, 10, 11, 14, 15)$$

PLA table:

[illegible]

Minimum SIZE of PLA is

Q7 Complete the following table with respect to the state of the respective transistor (ON or OFF) for the various input combinations and write the status of Output X (HIGH or LOW). [5]



Input A	Input B	Q3	Q4	Q5	Q6	Output X
0	0					
0	1					
1	0					
1	1					

Q8 For the following sequential circuit, construct the state table. [5]

State	Present State		Input X	Output Z	Flip-flop Input		Next State	
	Q ₂	Q ₁			J ₂ K ₂	J ₁ K ₁		
S ₀	0	0	0					
	0	0	1					
S ₁	0	1	0					
	0	1	1					
S ₂	1	0	0					
	1	0	1					
S ₃	1	1	0					
	1	1	1					

